# Tomas Herrera

Research Scientist **|** Machine Learning & E-Commerce Analytics

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## PROFESSIONAL EXPERIENCE

**Research Scientist, Quantix Labs |** Denver, CO **|** January 2024 – Present

Conducted research on recommendation systems and user behavior modeling in e-commerce contexts, with focus on neural architectures for product ranking. Developed and evaluated multiple sequence-to-sequence models on proprietary e-commerce datasets, achieving 12% improvement over baseline collaborative filtering approaches on held-out test sets. Published findings on attention mechanisms for fashion recommendation in the ICML 2025 Workshop on Machine Learning for E-Commerce (currently at 8 citations). Implemented benchmark suite for evaluating recommendation quality using standard metrics including NDCG@10 and MAP across multiple product categories. Collaborated with the academic partnerships team to define evaluation methodologies and dataset construction practices, though did not have operational responsibility for live systems.

**PhD Candidate in Machine Learning, University of Colorado Boulder |** Boulder, CO **|** August 2021 – December 2023

Completed doctoral research on deep learning approaches for product discovery and cross-category purchase prediction. Dissertation focused on novel transformer-based architectures designed to model sequential purchasing behavior in large retail catalogs. Conducted extensive experiments on three large-scale e-commerce datasets (Instacart, Amazon Reviews, and internal university dataset), reporting detailed ablation studies and performance metrics. Published primary dissertation work as "Sequential Embeddings for E-Commerce Product Graphs" in ACM Transactions on Information Systems (July 2023, 14 citations to date). Co-authored secondary publication on comparative analysis of representation learning methods in "Proceedings of the KDD 2023 Workshop on Learning on Graphs," which has accumulated 6 citations. Developed custom evaluation framework for measuring recommendation diversity and coverage on sparse product categories, establishing benchmarks that informed subsequent research directions.

**Graduate Research Assistant, University of Colorado Boulder |** Boulder, CO **|** August 2020 – July 2021

Assisted PhD advisor with research on graph neural networks applied to e-commerce product hierarchies. Conducted literature review and implemented baseline models for product classification tasks on academic benchmark datasets. Participated in weekly lab meetings and contributed to methodology discussions for large-scale recommendation system evaluation.

**Data Analysis Intern, RetailTech Innovations |** Fort Collins, CO **|** June 2019 – August 2019

Analyzed customer purchase patterns using SQL and Python to support research into seasonal demand forecasting. Prepared summary statistics and visualizations for presentation to research team.

## EDUCATION

**PhD in Computer Science**, University of Colorado Boulder **|** 2023

Dissertation: Sequential Embeddings for E-Commerce Product Graphs

**BS in Computer Science**, Colorado State University **|** 2019

## PUBLICATIONS & BENCHMARKS

- "Sequential Embeddings for E-Commerce Product Graphs," ACM Transactions on Information Systems, July 2023 (14 citations)
- "Attention Mechanisms for Fashion Recommendation," ICML 2025 Workshop on Machine Learning for E-Commerce (8 citations)
- "Comparative Analysis of Representation Learning Methods," KDD 2023 Workshop on Learning on Graphs (6 citations)

- Developed NDCG@10 and MAP evaluation benchmarks for e-commerce datasets with public reporting of results

## TECHNICAL SKILLS

Python, PyTorch, TensorFlow, SQL, Git, Jupyter, Scikit-learn, Transformers, GNNs, Recommendation Systems